

Skew monoidal categories and bialgebroids

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- ▶ Then we will look at skew monoidal categories and examples.
- ▶ Categories themselves and bialgebroids will be characterized as closed skew monoidal structures.
- ▶ We will conclude by discussing the free skew monoidal category and ‘coherence’.

Herds

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- ▶ More precisely, a *herd* A is a set with a ternary operation

$$q : A \times A \times A \longrightarrow A$$

satisfying

$$\begin{aligned} q(a, b, q(c, d, e)) &= q(q(a, b, c), d, e) , \\ q(a, b, b) &= a \text{ and } b = q(a, a, b) . \end{aligned}$$

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- ▶ Each group G is a herd with $q(a, b, c) = ab^{-1}c$.
- ▶ For any herd, choice of any element $j \in A$ makes A a group with multiplication $ab = q(a, j, b)$, identity element j and $a^{-1} = q(j, a, j)$.
- ▶ Let $\text{Iso}_{\mathcal{C}}(X, Y)$ be the set of invertible morphisms $a : X \longrightarrow Y$ in any category $\mathcal{C}$. It becomes a herd via $q(a, b, c) = ab^{-1}c$.
Choice of $j \in \text{Iso}_{\mathcal{C}}(X, Y)$ gives a group isomorphic to the group $\text{Aut}_{\mathcal{C}}(X)$ of automorphisms of $X \in \mathcal{C}$.

Flocks

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- ▶ More precisely, a *right flock* $\mathcal{A}$ is a category with a functor

$$Q : \mathcal{A} \times \mathcal{A}^{\text{op}} \times \mathcal{A} \longrightarrow \mathcal{A}$$

equipped with natural transformations

$$\phi_{ABCDE} : Q(A, B, Q(C, D, E)) \longrightarrow Q(Q(A, B, C), D, E)$$

$$\xi_A^B : Q(A, B, B) \longrightarrow A$$

$$\zeta_B^A : B \longrightarrow Q(A, A, B) ,$$

such that the following five conditions hold:

$$\begin{array}{ccc}
 Q(Q(A, B, C), D, Q(E, F, G)) & \xleftarrow{\phi} & Q(A, B, Q(C, D, Q(E, F, G))) \\
 \phi \downarrow & & \downarrow Q(1, 1, \phi) \\
 Q(Q(Q(A, B, C), D, E), F, G) & & \\
 \uparrow Q(\phi, 1, 1) & & \\
 Q(Q(A, B, Q(C, D, E)), F, G) & \xleftarrow{\phi} & Q(A, B, Q(Q(C, D, E), F, G)) \\
 \\
 Q(A, B, Q(B, B, C)) & \xrightarrow{\phi} & Q(Q(A, B, B), B, C) \\
 \uparrow Q(1, 1, \zeta) & & \downarrow Q(\xi, 1, 1) \\
 Q(A, B, C) & \xrightarrow{1} & Q(A, B, C)
 \end{array}$$

$$\begin{array}{ccc}
 Q(A, A, Q(B, C, D)) & \xrightarrow{\phi} & Q(Q(A, A, B), C, D) \\
 & \swarrow \zeta \quad \searrow Q(\zeta, 1, 1) & \\
 & Q(B, C, D) &
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$$\begin{array}{ccc}
 Q(A, B, Q(C, D, D)) & \xrightarrow{\phi} & Q(Q(A, B, C), D, D) \\
 & \searrow Q(1, 1, \xi) \quad \swarrow \xi & \\
 & Q(A, B, C) &
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$$\begin{array}{ccc}
 & Q(A, A, A) & \\
 \zeta \nearrow & & \searrow \xi \\
 A & \xrightarrow{1} & A \cdot \square
 \end{array}$$

Example

Consider the category $\mathbf{Vect}$ of finite dimensional (right) vector spaces over some fixed field k . For $V \in \mathbf{Vect}$, write V^* for the vector space of linear functions $\ell : V \longrightarrow k$. We have linear functions $\varepsilon : V^* \otimes V \longrightarrow k$ and $\eta : k \longrightarrow V \otimes V^*$ defined by $\varepsilon(\ell \otimes v) = \ell(v)$ and $\eta(1) = \sum_{i=1}^n e_i \otimes e_i^*$ where $\sum_{i=1}^n e_i e_i^*(v) = v$. Put

$$Q(U, V, W) = U \otimes V^* \otimes W .$$

Take ϕ to be the identity function of $U \otimes V^* \otimes W \otimes X^* \otimes Y$. Take

$$\xi = 1_U \otimes \varepsilon : U \otimes V^* \otimes V \longrightarrow U$$

and

$$\zeta = \eta \otimes 1_V : V \longrightarrow U \otimes U^* \otimes V .$$

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A flock with ϕ invertible is called *Hopf*.

Example

We make the category $\mathbf{Ab}^R$ of left R -modules into a flock. Put

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We have the canonical left R -module morphism

$$\tau : \mathrm{Hom}(N, M) \otimes_R \mathrm{Hom}(L, K) \longrightarrow \mathrm{Hom}(\mathrm{Hom}_R(\mathrm{Hom}(N, M), L), K)$$

defined by $\tau(f \otimes_R g)(\theta) = g(\theta(f))$. Applying the contravariant functor $\mathrm{Hom}_R(-, H)$ to τ and using the $\otimes_R$ - Hom_R adjunction, we obtain our

$$\begin{aligned} \phi : \mathrm{Hom}_R(\mathrm{Hom}(\mathrm{Hom}_R(\mathrm{Hom}(N, M), L), K), H) \\ \longrightarrow \mathrm{Hom}_R(\mathrm{Hom}(N, M), \mathrm{Hom}_R(\mathrm{Hom}(L, K), H)) . \end{aligned}$$

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Moreover, $\xi : \mathrm{Hom}_R(\mathrm{Hom}(M, M), L) \longrightarrow L$ evaluates at 1_M and $\zeta : L \longrightarrow \mathrm{Hom}_R(\mathrm{Hom}(M, L), L)$ is $\zeta(x)(\theta) = \theta(x)$.

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Take $\xi : M \otimes \text{Hom}_R(M, L) \longrightarrow L$ to be evaluation and

$\zeta : M \longrightarrow M \otimes \text{Hom}_R(L, L)$ to be $\zeta(x) = x \otimes 1_L$.

Skew monoidal categories

A category $\mathcal{R}$ is *right skew-monoidal* when it is equipped with

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- ▶ natural families of (not necessarily invertible) morphisms

$$\alpha : A \otimes (B \otimes C) \longrightarrow (A \otimes B) \otimes C$$

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- ▶ plus five axioms as follows:

The pentagon and trivial unit condition

$$\begin{array}{ccc}
 & (W \otimes X) \otimes (Y \otimes Z) & \\
 \alpha_{W,X,Y \otimes Z} \nearrow & & \searrow \alpha_{W \otimes X,Y,Z} \\
 W \otimes (X \otimes (Y \otimes Z)) & & ((W \otimes X) \otimes Y) \otimes Z \\
 \downarrow 1_W \otimes \alpha_{X,Y,Z} & & \uparrow \alpha_{W,X,Y} \otimes 1_Z \\
 W \otimes ((X \otimes Y) \otimes Z) & \xrightarrow{\alpha_{W,X \otimes Y,Z}} & (W \otimes (X \otimes Y)) \otimes Z
 \end{array}
 \quad (1)$$

$$\begin{array}{ccc}
 I & \xrightarrow{1_I} & I \\
 \searrow \lambda_I & & \nearrow \rho_I \\
 & I \otimes I &
 \end{array}
 \quad (2)$$

Three more unit conditions

$$\begin{array}{ccc}
 I \otimes (X \otimes Y) & \xrightarrow{\alpha_{I,X,Y}} & (I \otimes X) \otimes Y \\
 & \nwarrow \lambda_{X \otimes Y} \quad \nearrow \lambda_X \otimes 1_Y & \\
 & X \otimes Y &
 \end{array} \tag{3}$$

$$\begin{array}{ccc}
 X \otimes (I \otimes Y) & \xrightarrow{\alpha_{X,I,Y}} & (X \otimes I) \otimes Y \\
 1_X \otimes \lambda_Y \uparrow & & \downarrow \rho_X \otimes 1_Y \\
 X \otimes Y & \xrightarrow{1_{X \otimes Y}} & X \otimes Y
 \end{array} \tag{4}$$

$$\begin{array}{ccc}
 X \otimes (Y \otimes I) & \xrightarrow{\alpha_{X,Y,I}} & (X \otimes Y) \otimes I \\
 & \searrow 1_X \otimes \rho_Y \quad \swarrow \rho_{X \otimes Y} & \\
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- ▶ $\lambda : A \longrightarrow J \otimes A$ is $\zeta : A \longrightarrow Q(J, J, A)$.

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- ▶ $\lambda : A \longrightarrow J \otimes A$ is $\zeta : A \longrightarrow Q(J, J, A)$.
- ▶ $\rho : A \otimes J \longrightarrow A$ is $\xi : Q(A, J, J) \longrightarrow A$.

Example from a bialgebra

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- ▶ $\lambda : A \longrightarrow H \otimes A$ is $\eta \otimes 1_A$ and $\rho : H \otimes A \longrightarrow A$ is $\varepsilon \otimes 1_A$.
- ▶ This skew monoidal structure on $\mathbf{Vect}$ is Hopf if and only if the bialgebra H is a Hopf algebra.

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- ▶ The unit object J is defined by $J_x = 1$ for all $x \in X$.
- ▶ Observe that

$$(M *_A (N *_A L))_x = \sum_{y,z} M_x \times A(x, y) \times N_y \times A(y, z) \times L_z$$

$$((M *_A N) *_A L)_x = \sum_{y,z} M_x \times A(x, y) \times N_y \times A(x, z) \times L_z$$

Example from a category (continued)

- ▶ The morphism $\alpha : M *_A (N *_A L) \longrightarrow (M *_A N) *_A L$ uses the function

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- ▶ The morphism $\rho : M *_A J \longrightarrow M$ projects $\sum_y M_x \times A(x, y) \longrightarrow M_x$.

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$$N_x \xrightarrow{\text{id}_x \times 1} A(x, x) \times N_x \xrightarrow{\text{in}_x} \sum_y A(x, y) \times N_x .$$

- ▶ The morphism $\rho : M *_A J \longrightarrow M$ projects $\sum_y M_x \times A(x, y) \longrightarrow M_x$.
- ▶ This skew monoidal structure on Set^X is Hopf iff A is a groupoid.

Closed skew monoidal categories

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Theorem

Category structures with object set X are equivalent to closed right-skew monoidal structures on the category Set^X of X -indexed families of sets for which the skew unit is the terminal object.

Bialgebroids

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- ▶ If you know what bialgebroids are, the following will be a Theorem. Otherwise, it can be a Definition.

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▶ Theorem (Szlachányi 2012)

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- ▶ The proof begins by noting that tensor products

$$\mathbf{Ab}^{R^\circ} \otimes \mathbf{Ab}^{R^\circ} \longrightarrow \mathbf{Ab}^{R^\circ},$$

which are colimit-preserving in each variable, are induced by tensoring with a left $R \otimes R$ -, right- R -bimodule A . This A becomes a ring with two ring morphisms $s : R^\circ \longrightarrow A$ and $t : R \longrightarrow A$ with commuting images.

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- ▶ Let $\Delta_{\top}$ denote the category whose objects are the finite ordinals $\mathbf{n} = \{0, 1, \dots, n-1\}$ with last element (so that $n > 0$), and whose morphisms are last-element-and-order-preserving functions.

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- ▶ $\Delta_{\top}$ is Hopf left skew monoidal with ordinal sum as tensor product, with unit $\mathbf{1}$, with right unit constraint $\rho : \mathbf{n} + \mathbf{1} \longrightarrow \mathbf{n}$ the surjection identifying $n-1$ and n , and with left unit constraint $\lambda : \mathbf{m} \longrightarrow \mathbf{1} + \mathbf{m}$ the injection whose image does not contain 0.

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Theorem (Lack-Street 2013)

*The unique strict monoidal functor **Fsk** $\longrightarrow$ $\Delta_{\top}$, which takes the generating object **N** of **Fsk** to the object **1** of $\Delta_{\top}$, is faithful.*